Chapter 6

Representation of Finite Groups

6.1 Representations of Finite Groups

Definition 6.1. Let V be a vector space. The **general linear group** of V is

$$GL(V) = \{L : V \to V \mid L \text{ is a bijective linear transformation}\}$$

Remark
$GL(V)$ is a group. Group operation is composition of functions, identity is the identity map, and inverse is the inverse of function.

Definition 6.2. Let G be a finite group. A **representation** (a.k.a linear representation) of G is a group homomorphism $\rho : G \to GL(V)$ where V is a finite-dimensional vector space over $\mathbb{C}$. Define the degree of ρ to be the dimension of V.

Remark
A representation $\rho : G \to GL(V)$ induced an action of G on V by $g \cdot v = \rho(g)(v)$ with

- $g \cdot (\alpha v + \beta w) = \alpha(g \cdot v) + \beta(g \cdot w)$ (linear transformation of $\rho(g)$
- $(gh) \cdot v = g \cdot (h \cdot v)$ (composition)
- $e_G \cdot v = v$ (identity)

We may write $\rho(g)(v)$ as gv for simplicity.

Definition 6.3. Let $\rho : G \to GL(V)$ be a representation of a group G. If in addition V has an inner product $\langle \cdot, \cdot \rangle$ s.t.

$$\langle \rho(g)(v), \rho(g)w \rangle = \langle gv, gw \rangle = \langle v, w \rangle$$

for all $g \in G$ and $v, w \in V$, then we call ρ a **unitary** representation with respect to $\langle \cdot, \cdot \rangle$. Consider ρ as an action of G on V, we also say $\langle \cdot, \cdot \rangle$ is $G-invariant$.

Remark
Recall definition for a unitary operator (linear transformation):

$$\langle Tv, Tw \rangle = \langle v, w \rangle$$

for all $v, w \in V$. Also recall that the matrix representation of the operator under the same inner product must be unitary.

Remark
The trivial representation $\phi : G \to GL(\mathbb{C})$ by $\phi(g)\alpha = \alpha$ for all $g \in G$ and $\alpha \in \mathbb{C}$ is unitary with respect to $\langle \cdot, \cdot \rangle_2$, the standard inner product on $\mathbb{C}$ given by $\langle \alpha, \beta \rangle_2 = \alpha\overline{\beta}$. Since

$$\langle \phi(g)\alpha, \phi(g)\beta \rangle_2 = \langle \alpha, \beta \rangle_2$$

Definition 6.4. Let G be a finite group. Recall that

$$L^2(G) = \{f : G \to \mathbb{C}\}$$

The **right regular representation**, $R : G \to GL(L^2(G))$, is defined by

$$R(\gamma)f(g) = f(g\gamma)$$

for all $f \in L^2(G)$ and $\gamma, g \in G$

Example
Consider $f \in L^2(\mathbb{Z}_4)$ by

$$f(n) = \begin{cases} 1 & n = 0 \\ 0 & n = 1 \\ 1 - 15i & n = 2 \\ -2/3 & n = 3 \end{cases}$$

Then $R(1)f(n) = f(1n) = f(n+1)$. Hence,

$$R(1)f(n) = \begin{cases} 0 & n = 0 \\ 1 - 15i & n = 1 \\ -2/3 & n = 2 \\ 1 & n = 3 \end{cases}$$

Lemma 6.5.

- Properties of right regular representation

- R is a homomorphism
- R is unitary with respect to the standard inner product on $L^2(G)$
- Degree of R is $|G|$

Proof

- Let $\gamma, \gamma', g \in G$ and $f \in L^2(G)$. Then
$$R(\gamma\gamma')f(g) = f(g\gamma\gamma')$$
while
$$R(\gamma)[R(\gamma')f](g) = R(\gamma')f(g\gamma) = f(g\gamma\gamma')$$
- Let $f, h \in L^2(G)$ and $\gamma \in G$, then
$$\begin{aligned}\langle R(\gamma)f, R(\gamma)h\rangle_2 &= \sum_{g\in G} R(\gamma)f(g)\overline{R(\gamma)h(g)} \\ &= \sum_{g\in G} f(g\gamma)\overline{h(g\gamma} \\ &= \sum_{x\in G} f(x)\overline{h(x)} \\ &\text{group is closed, just a permutation} \\ &= \langle f, h\rangle_2\end{aligned}$$
- Recall the standard basis remark for Def. 1.10, the dimension of $L^2(G)$ is $|G|$.

Definition 6.6. Let G be a finite group. A **matrix representation** of G is a homomorphism $\pi : G \to GL(n, \mathbb{C})$. The degree of π is n. We say π is a unitary matrix representation if $\pi(g)$ is a unitary matrix for all $g \in G$.

Lemma 6.7. Let G be a finite group. One can go back and forth between two different definitions of group representations as follows.

- Let $\pi : G \to GL(n, \mathbb{C})$ be a matrix representation of G of degree n. Then the map $\rho : G \to GL(\mathbb{C}^n)$ defined by $\rho(g)(v) = \pi(g)v$ is a representation of G of degree n.
- Let $\rho : G \to GL(V)$ be a representation of degree n. Fix an ordered basis β for V. We can define a matrix representation $\pi : G \to GL(n, \mathbb{C})$ by $\pi(g) = [\rho(g)]_\beta$. The degree of π is n. We can check that π is indeed a homomorphism.

- Suppose we chose a different basis β'. Then, there exists an invertible matrix Q s.t. $[\rho(g)]_{\beta'} = Q^{-1}[\rho(g)]_{\beta}Q$. That is, $[\rho(g)]_{\beta}$ and $[\rho(g)]_{\beta'}$ are similar matrices.

Proof
This is essentially linear algebra. Ordered basis is used to write vectors in coordinate form. Write $[v]_\beta$ for vector v expressed as coordinate form by ordered basis β.

- In general, for linear transformation $L : V \to W$, with ordered basis $\beta = [v_1, v_2, \dots, v_n]$ for V and γ for W, we have the corresponding matrix of L with respect to β and γ as

$$[L]_\beta^\gamma = \Big([L(v_1)]_\gamma \;\Big|\; [L(v_2)]_\gamma \;\Big|\; \dots \;\Big|\; [L(v_n)]_\gamma\Big)$$

  If $V = W$ and $\beta = \gamma$, we write as $[L]_\beta$

- Intuitively, you first write vector as coordinate by basis β, then the matrix tells you how much value you get under coordinate γ for 1 unit under coordinate β after transformation L, and you multiple this ratio to each entry of coordinate of the vector under β, done. That is,

$$[L(v)]_\gamma = [L]_\beta^\gamma [v]_\beta = \Big([L(v_1)]_\gamma \;\Big|\; [L(v_2)]_\gamma \;\Big|\; \dots \;\Big|\; [L(v_n)]_\gamma\Big)(a_1, a_2, \dots, a_n)_\beta^T$$

- Note in general the changes in coordinate come from both the change of basis and the linear transformation. By making L as the identity transformation I (hence no change), we have a matrix that changes basis only. That is

$$[v]_\gamma = [I(v)]_\gamma = [I]_\beta^\gamma [v]_\beta$$

- Denote $Q = [I]_\beta^\gamma$, we know:
  - Q is invertible and $Q^{-1} = [I]_\gamma^\beta$
  - For any $v \in V$, $[v]_\gamma = Q[v]_\beta$
  - $[L]_\beta = Q^{-1}[L]_\gamma Q$

- Homomorphism of π from linear transformation

Definition 6.8. Let $\rho : G \to GL(V)$ and $\phi : G \to GL(W)$ be two representations of G. Let $\theta : V \to W$ be a map.

- If $\theta(\rho(g)v) = \phi(g)\theta(v)$ for all $g \in G$ and $v \in V$, then θ is $G-invariant$
- A $G-homomorphism$ is a G-invariant linear transformation
- If in addition θ is a vector space isomorphism, then we say V and W are equivalent and write $V \cong W$, $\rho \cong \phi$

Remark
G-invariant means θ commutes with the action of G, so it does not matter whether apply θ first or apply G-action first. In Def. 6.3 we discussed a special case of inner product.

Remark
Let $\rho : G \to GL(V)$ and $\phi : G \to GL(W)$ be two representations of G. Let $\theta : V \to W$ be G-homomorphism. Let β be an ordered basis for V and γ be an ordered basis for W. Then, translating Def. 6.8 into matrices,

$$[\theta]_\beta^\gamma [\rho(g)]_\beta [v]_\beta = [\phi(g)]_\gamma [\theta]_\beta^\gamma [v]_\beta$$

for all $v \in V$ and $g \in G$. Thus

$$[\theta]_\beta^\gamma [\rho(g)]_\beta = [\phi(g)]_\gamma [\theta]_\beta^\gamma$$

Moreover, if θ is an isomorphism (hence corresponding matrix is invertible), then

$$[\theta]_\beta^\gamma [\rho(g)]_\beta ([\theta]_\beta^\gamma)^{-1} = [\phi(g)]_\gamma$$

Definition 6.9. Let G be a finite group and $\phi, \pi : G \to GL(n, \mathbb{C})$ be two matrix representations of G. We say that ϕ and π are equivalent if there is a matrix $Q \in GL(n, \mathbb{C})$ s.t. $Q\phi(g)Q^{-1} = \pi(g)$ for all $g \in G$, and write $\phi \cong \pi$. That is, for the same element in G, we always get similar matrices under two matrix representations.

Remark
In

$$[\theta]_\beta^\gamma [\rho(g)]_\beta ([\theta]_\beta^\gamma)^{-1} = [\phi(g)]_\gamma$$

Let $\theta = I$, where I is the identity map (hence identity matrix). Then we have $\rho \cong \phi$ from Def.6.8. Denote $[I]_\beta^\gamma = Q$, the change of basis matrix. Then,

$$Q[\rho(g)]_\beta Q^{-1} = [\phi(g)]_\gamma$$

Again denote $[\rho(g)]_\beta = \pi(g), [\phi(g)]_\gamma = \phi(g)$, i.e. matrix corresponding to linear transformations. Then,

$$Q\pi(g)Q^{-1} = \phi(g)$$

That is, equivalent matrix representations of G can be thought of as arising from equivalent (the same) representation of G.

6.2 Decomposing into Irreducible Representations

Next section will show that these decomposition is unique.

Definition 6.10.

- Let G be a finite group and $\rho : G \to GL(V)$ be a representation of G. We say that a subspace W of V is a $G - invariantsubspace$, or $subrepresentation$ of V, if $\rho(g)(w) \in W$ for all $g \in G, w \in W$
- The G-invariant subspace $\{0\}$ and V are trivial subrepresentations of V
- V is **reducible** if it contains a nontrivial G-invariant subspace W. Otherwise, we say that V is **irreducible**, or V is an **irrep**

Remark
We should distinguish "the" trivial representation (one dimensional representation that sends every $g \in G$ to identity) and trivial representations (representations that sends every $g \in G$ to the identity map of some space V). Clearly, "the" trivial representation is a trivial representation, and it is irreducible (since $\mathbb{C}$ do not have nontrivial subspaces / subrepresentations). All other trivial representations that is not equivalent to "the" trivial representation is reducible (any subspace would be invariant, and they all have nontrivial subspaces).

Remark
Let $\rho(g)|_W$ denotes the restriction of $\rho(g)$ to the subspace $W \subset V$, we have $\rho(g)|_W : W \to W$. Hence we can define a new representation $\rho_W : G \to GL(W)$ by $\rho_W(g) = \rho(g)|_W$

Remark
Suppose that W and W' are subrepresentations of V and that $V = W \bigoplus W'$. Let $v \in V$, then $v = w + w'$ for some uniquely determined $w \in W, w' \in W'$. Let $\rho, \rho_W, \rho_{W'}$ be defined respectively, then

$$\rho(g)(v) = \rho(g)(w) + \rho(g)(w') = \rho_W(g)(w) + \rho_{W'}(g)(w')$$

Definition 6.11. Under the conditions of previous remark, we say that ρ is the **direct sum** of $\rho_W, \rho_{W'}$, and we write $\rho = \rho_W \bigoplus \rho_{W'}$. We also write $n\pi$ for π being direct summed n times.

Definition 6.12. Let X, Y, Z be matrices. We say $X = Y \bigoplus Z$ if

$$X = \begin{pmatrix} Y & 0 \\ 0 & Z \end{pmatrix}$$

We also write nX for X being direct summed n times.

Definition 6.13.

- Given two matrix representations π_1 and π_2 of a group G, define the **direct sum** of π_1 and π_2, denoted $\pi_1 \bigoplus \pi_2$, as the matrix representation of G given by $(\pi_1 \bigoplus \pi_2)(g) = \pi_1(g) \bigoplus \pi_2(g)$. Write $n\pi$ for π being direct summed n times.
- Let $\phi : G \to GL(n, \mathbb{C})$ be a matrix representation for a finite group G. We say that ϕ is reducible if $\phi \cong \pi_1 \bigoplus \pi_2$ for some matrix representations π_1, π_2 of G. Otherwise, ϕ is **irreducible**, or an **irrep**.

Remark
Let $\rho : G \to GL(V)$ be a representation of a finite group G, and $V = V_1 \bigoplus V_2 \bigoplus \cdots \bigoplus V_n$, where each V_i is a G-invariant subspace of V with respect to ρ. Let β_i be a basis for V_i, $\beta = \bigcup_{i=1}^{n} \beta_i$ and $\rho_i(g) = \rho(g)|_{V_i}$, then

$$[\rho(g)]_\beta = \begin{pmatrix} [\rho_1(g)]_{\beta_1} & 0 & \dots & 0 \\ 0 & [\rho_2(g)]_{\beta_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & [\rho_n(g)]_{\beta_n} \end{pmatrix}$$

for every $g \in G$.

Lemma 6.14. Let $\rho : G \to GL(V)$ be a representation of a finite group G. Then there exists a G-invariant inner product on V.

Proof
Let $\beta = [v_1, v_2, \dots, v_n]$ be an arbitrary ordered basis of V. Define an inner product on V following the basis as below. Given

$$\begin{aligned} v &= a_1 v_1 + a_2 v_2 + \cdots + a_n v_n \\ w &= b_1 w_1 + b_2 w_2 + \cdots + b_n w_n \end{aligned}$$

Define

$$\langle v, w \rangle = a_1 \overline{b_1} + a_2 \overline{b_2} + \cdots + a_n \overline{b_n}$$

This is indeed an inner product. Bilinear, conjugate symmetry, and positive definite are all obvious. However, ρ may not be unitary with respect to this inner product. Define another inner product $\langle \rangle'$

$$\langle v, w \rangle' = \sum_{g \in G} \langle \rho(g) v, \rho(g) w \rangle$$

Then ρ is unitary with respect to $\langle \rangle'$

$$\begin{aligned} \langle \rho(t) v, \rho(t) w \rangle' &= \sum_{g \in G} \langle \rho(g)\rho(t) v, \rho(g)\rho(t) w \rangle \\ &= \sum_{x \in G} \end{aligned}$$

Theorem 6.15. Maschke's Theorem
Every unitary representation of a finite group can be completely decomposed into orthogonal irreducible representations.
Let V be a unitary representation of G with a G-invariant $\langle \cdot, \cdot \rangle$. Then there exists irreducible subrepresentations $V_1, \ldots, V_n$ of V s.t.

$$V = V_1 \bigoplus V_2 \bigoplus \cdots \bigoplus V_n$$

Remark
Equivalently, (after combining copies of identical representations (subrepresentations) this is also to say:
Given unitary representation ρ (V) of G with a G-invariant $\langle , \rangle$, there exists irreducible representations $\rho_1, \ldots, \rho_n$ (subrepresentations $V_1, \ldots, V_n \in V$) s.t.

$$\rho = a_1\rho_1 \bigoplus \cdots \bigoplus a_n\rho_n$$
$$V = a_1V_1 \bigoplus \cdots \bigoplus a_nV_n$$

Proof
Proof idea: if V is not irreducible, then decompose it as the direct sum of a subrepresentation and its orthogonal complement to reduce the dimension. Show the complement is also a subrepresentation, then induction follows.

- For nontrivial V, $V = W \bigoplus W^{\perp}$
- Key step: given W as a nontrivial G-invariant subspace of V, its orthogonal complement $W^{\perp}$ is also a subrepresentation.

Corollary 6.16. Let $\rho : G \to GL(n, \mathbb{C})$ be a unitary matrix representation of a finite group G, then there is a unitary matrix $Q \in GL(n, \mathbb{C})$ s.t.

$$Q^{-1}\rho(g)Q = \begin{pmatrix} [\rho_1(g)] & 0 & \ldots & 0 \\ 0 & [\rho_2(g)] & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & [\rho_n(g)] \end{pmatrix}$$

for every $g \in G$, where each ρ_i is an irreducible matrix representation of G.

Remark
Q here is essentially the change of basis matrix. The statement is just translating Maschke's theorem (linear transformation form) into matrix form.

Lemma 6.17. Any matrix representation of G is equivalent to a unitary matrix representation of G.

Proof

- Assume $dim(V) = n$. Suppose $\rho : G \to GL(V)$ is a unitary representation with respect to some inner product $\langle , \rangle$. Let $\beta = [v_1, \dots, v_n]$ be an ordered orthonormal basis for V with respect to $\langle , \rangle$ (existence guaranteed by Gram-Schmidt process). Easy to check that $[\rho(g)]_\beta$ is a unitary matrix for all $g \in G$.

- Suppose $\pi : G \to GL(n, \mathbb{C})$ is a matrix representation of G. Then, by Lemma 6.7, a representation $\rho : G \to GL(\mathbb{C}^n)$ is defined. Take the inner product that makes ρ unitary as the inner product of $GL(n, \mathbb{C})$. By previous argument, this representation has a corresponding basis s.t. $[\rho(g)]_\beta$ is unitary to for every $g \in G$. Then, there exists a change-of-basis matrix Q s.t. the matrix representation $\phi(g) = T^{-1}\pi(g)T$ is unitary. That is, simply change the basis for π (for every $\pi(g)$) and we get a unitary matrix representation that is equivalent to π.

- The key here is that the inner product of $\mathbb{C}^n$ need not to coincidence with the inner product that makes ρ unitary. $[\rho(g)]_\beta$ will always be a unitary matrix whatever the inner product of the space is.

6.3 Schur's Lemma and Characters of Representation

Definition 6.18. Let G be a finite group.

- Let $\pi : G \to GL(n, \mathbb{C})$ be a matrix representation. The character of π, denoted χ_π, is $\chi_\pi(g) = \text{tr}(\pi(g))$ for all $g \in G$. (tr() means trace of a matrix)

- Let $\rho : G \to GL(V)$ be a representation of G. The character of ρ, denoted χ_ρ, is $\chi_\rho(g) = \text{tr}([\rho(g)]_\beta)$, where β is any basis of V.

- Suppose χ is the character of a representation (matrix representation) ρ on G, then degree of χ is $\chi(e_G)$.

- A character χ is an irreducible character if there exists an irreducible representation ρ s.t. $\chi = \chi_\rho$.

Remark

- Character is a function, but with $G \to \mathbb{R}$

- Similar matrices have the same trace (since trace is the sum of all eigenvalues), so the character of representation ρ is independent of the choice of basis.

- Since e_G is always mapped to the identity matrix (representation must be a homomorphism), for $\rho : G \to GL(V)$, $\chi(e_G) = dim(V)$.

- In Corollary 6.27 we see that two representations are equivalent iff they have the same character. Hence, we can say χ_ρ is irreducible iff ρ is irreducible.

Definition 6.19. Group G, $g, h \in G$. g, h are conjuage in G if $g = xhx^{-1}$ for some $x \in G$. The conjugacy class of g is

$$K_g = \{xgx^{-1} \mid x \in G\}$$

Lemma 6.20. Finite group G, representations (matrix representations) ϕ, π of G with characters χ, ψ. Then

- If g, h conjugate in G, then $\chi(g) = \chi(h)$. That is, characters are constant on conjugacy classes.
- If $\phi \cong \pi$, then $\chi = \psi$

Remark
To define a character, suffice to define it for every conjugacy class of G.

Proof
We prove for matrix representations of G, and the proof extends to representations by choosing bases.

- Suppose $g = xhx^{-1}$ for some $x \in G$. Then

$$\begin{aligned}\chi(g) = \operatorname{tr}(\phi(g)) &= \operatorname{tr}(\phi(x)\phi(h)\phi(x)^{-1}) \\ &\text{homomorphism} \\ &= \operatorname{tr}(\phi(h)) = \chi(h) \\ &\text{similar matrix same trace}\end{aligned}$$

- If $\phi \cong \pi$, then $\phi(g) = X\pi(g)X^{-1}$ for all $g \in G$. Then

$$\chi(g) = \operatorname{tr}(\phi(g)) = \operatorname{tr}(X\pi(g)X^{-1}) = \operatorname{tr}(\pi(g)) = \psi(g)$$

Theorem 6.21. (Schur's Lemma)
Let V, W be irreducible representations of G, let $\theta : V \to W$ be a G-homomorphism. Then either θ is the zero map or θ is an isomorphism, in which case two representations are equivalent.

Proof

- Let $ker(\theta)$ be the kernel, $im(\theta)$ be the image. Then clearly $ker(\theta) \subset V$ and $im(\theta) \subset W$.
- Let $v \in ker(\theta), g \in G$. Then $\theta(g \cdot v) = g \cdot \theta(v)$ (G-invariant) $= g \cdot 0 = 0$, so $g \cdot v \in ker(\theta)$, thus $ker(\theta)$ is a subrepresentation of V.
- Let $\theta(v) \in im(\theta), g \in G$. Then $g \cdot \theta(v) = \theta(g \cdot v) \in im(\theta)$, also a subrepresentation.
- Since V is irreducible, it only contains trivial subrepresentations, therefore either $ker(\theta) = V$ or $ker(\theta) = \{0\}$. Same argument for $im(\theta)$. Because θ is also a linear transformation, put everything together, we either have $ker(\theta) = V$ and $im(\theta) = \{0\}$, or $ker(\theta) = \{0\}$ and $im(\theta) = V$. Thus, θ is either the zero map or else an isomorphism.

Corollary 6.22. (Matrix version Schur's Lemma)
Let $\pi : G \to GL(n, \mathbb{C})$ and $\phi : G \to GL(m, \mathbb{C})$ be irreducible matrix representations of G, suppose that Z is an $n \times m$ matrix and $\pi(g)Z = Z\phi(g)$ for all $g \in G$. Then either Z is the zero matrix or Z is an invertible matrix, in which case $\pi \cong \phi$.

Corollary 6.23. Let $\pi : G \to GL(n, \mathbb{C})$ be an irreducible matrix representation of G. A matrix $X \in GL(n, \mathbb{C})$ has $X\pi(g) = \pi(g)X$ for all $g \in G$ iff $X = cI$, where $c \in \mathbb{C}$ and I is the $n \times n$ identity matrix.

Proof
Suppose X has $X\pi(g) = \pi(g)X$ for all $g \in G$. Let c be an eigenvalue of X, then $det(X - cI) = 0$ and $X - cI$ not invertible. Also, we have $(X - cI)\pi(g) = \pi(g)(X - cI)$ as an implication. Then, by Corollary 6.22, $X - cI = 0$, done. Conversely, every matrix of form $X = cI$ commutes with $\pi(g)$ for all $g \in G$, done.

Lemma 6.24. Let $\pi : G \to GL(n, \mathbb{C})$ and $\phi : G \to GL(m, \mathbb{C})$ be irreducible unitary matrix representations of G. Denote entries of these matrices as $\pi_{i,j}(g)$ and $\phi_{r,s}(g)$. The coordinate functions $\pi_{i,j}, \phi_{r,s}$ are elements of $L^2(G)$, and they have following orthogonality relations with respect to the standard inner product on $L^2(G)$.

- The coordinate functions of inequivalent, irreducible unitary representations are orthogonal. For inequivalent π, ϕ
$$\langle \pi_{i,j}, \phi_{r,s} \rangle_2 = 0$$
- The coordinate functions of an irreducible unitary representation (or, equivalent representations) are pairwise orthogonal.
$$\langle \pi_{i,j}, \pi_{r,s} \rangle_2 = \begin{cases} |G|/n & \text{if } i = r \text{ and } j = s \\ 0 & \text{otherwise} \end{cases}$$

Remark
Note unitary matrix has nothing to do with inner product.

Proof

- Suppose inequivalent π, ϕ. Given arbitrary j, s, let $E_{j,s}$ be the $n \times m$ matrix with a 1 on (j, s) entry and 0 everywhere else. Define the $n \times m$ matrix Y

  $$Y = \sum_{g \in G} \pi(g) E_{j,s} \phi(g^{-1}) = \sum_{g \in G} \pi(g) E_{j,s} \overline{\phi(g)^T}$$

  where the second equality follows because $\phi(g)$ is a unitary matrix and so $\phi(g)\overline{\phi(g)^T} = I$
  For each $h \in G$, note

  $$\begin{aligned} \pi(h)Y &= \sum_{g \in G} \pi(hg) E_{j,s} \phi(g^{-1}) = \sum_{x \in G} \pi(x) E_{j,s} \phi(x^{-1}h) \\ &= \left(\sum_{x \in G} \pi(x) E_{j,s} \phi(x^{-1}) \right) \phi(h) = Y\phi(h) \end{aligned}$$

  where $x = hg$ and thus $g^{-1} = x^{-1}h$. Since π not equivalent to ϕ, then by Corollary 6.22 $Y = 0$. Now compute (matrix multiplication) Y

  $$0 = \sum_{g \in G} \pi(g) E_{j,s} \overline{\phi(g)^T} \quad = \sum_{g \in G} \begin{pmatrix} \pi_{1,j}(g)\overline{\phi_{1,s}(g)} & \cdots & \pi_{1,j}(g)\overline{\phi_{m,s}(g)} \\ \vdots & \ddots & \vdots \\ \pi_{n,j}(g)\overline{\phi_{1,s}(g)} & \cdots & \pi_{n,j}(g)\overline{\phi_{m,s}(g)} \end{pmatrix}$$

  Hence every entry of the ultimate matrix is 0. Also note by definition of standard inner product, each entry is just the result of an inner product

  $$\langle \pi_{i,j}, \phi_{r,s} \rangle_2 = \sum_{g \in G} \pi_{i,j}(g)\overline{\phi_{r,s}(g)} = 0$$

- Let $E_{j,s}$ be defined as above. Define

  $$Y = \sum_{g \in G} \pi(g) E_{j,s} \pi(g^{-1})$$

  As before, for each $h \in G$, we have $\pi(h)Y = Y\pi(h)$. By Corollary 6.22, since $\pi \cong \pi$, we must have $Y = cI$, where I is the $n \times n$ identity matrix.

Then $\mathrm{tr}(Y) = nc$, so

$$
\begin{aligned}
c = \frac{\mathrm{tr}(Y)}{n} &= \frac{1}{n}\mathrm{tr}(\sum_{g\in G} \pi(g)E_{j,s}\pi(g^{-1})) \\
&= \frac{1}{n}\sum_{g\in G} \mathrm{tr}(\pi(g)E_{j,s}\pi(g)^{-1}) \\
&\text{sum of trace is trace of sum, also } \pi \text{ is homomorphism} \\
&= \frac{1}{n}\sum_{g\in G} \mathrm{tr}(E_{j,s}) \\
&= \begin{cases} \frac{|G|}{n} & j = s \\ 0 & \text{otherwise} \end{cases}
\end{aligned}
$$

Same as part 1, we have

$$
cI = Y = \sum_{g\in G} \begin{pmatrix} \pi_{1,j}(g)\overline{\phi_{1,s}(g)} & \cdots & \pi_{1,j}(g)\overline{\phi_{m,s}(g)} \\ \vdots & \ddots & \vdots \\ \pi_{n,j}(g)\overline{\phi_{1,s}(g)} & \cdots & \pi_{n,j}(g)\overline{\phi_{m,s}(g)} \end{pmatrix}
$$

Hence

$$
\begin{aligned}
\langle \pi_{i,j}, \pi_{r,s}\rangle_2 &= \sum_{g\in G} \pi_{i,j}(g)\overline{\pi_{r,s}(g)} = Y_{i,r} = (cI)_{i,r} \\
&= \begin{cases} \frac{|G|}{n} & \text{if } i = r \text{ and } j = s \\ 0 & \text{otherwise} \end{cases}
\end{aligned}
$$

That is, only get value for diagonal entries.

Theorem 6.25. Let π and ϕ be irreducible representations (or matrix representations) of G with corresponding characters χ, ψ respectively. Then

$$
\langle \chi, \psi\rangle_2 = \begin{cases} |G| & \pi \cong \phi \\ 0 & \text{otherwise} \end{cases}
$$

Proof
Prove for matrix representations. The same proof works for representations by picking bases. By Lemma 6.17, since every matrix representation is equivalent to a unitary matrix representation (thus having same character), for discussion on characters, we can assume that π, ϕ are unitary.

- Suppose $\pi \cong \phi$, by Lemma 6.20 (2), $\chi = \psi$. Hence, by Lemma 6.24 (for last equality),

$$\begin{aligned}
\langle \chi, \psi \rangle_2 &= \langle \chi, \chi \rangle_2 \\
&= \sum_{g \in G} \chi(g)\overline{\chi(g)} = \sum_{g \in G} \mathrm{tr}(\pi(g))\overline{\mathrm{tr}(\pi(g))} \\
&= \sum_{g \in G} \sum_{a=1}^{n} \sum_{b=1}^{n} \pi_{a,a}(g)\overline{\pi_{b,b}(g)} \\
&= \sum_{a=1}^{n} \sum_{b=1}^{n} \sum_{g \in G} \pi_{a,a}(g)\overline{\pi_{b,b}(g)} \\
&= \sum_{a=1}^{n} \sum_{b=1}^{n} \langle \pi_{a,a}, \pi_{b,b} \rangle_2 \\
&= n \cdot \frac{|G|}{n} = |G|
\end{aligned}$$

- Suppose otherwise, then by Lemma 6.24 again

$$\langle \chi, \psi \rangle_2 = \sum_{g \in G} \chi(g)\overline{\psi(g)} = \sum_{a=1}^{n} \sum_{b=1}^{m} \langle \pi_{a,a}, \phi_{b,b} \rangle_2 = 0$$

Corollary 6.26. Let π be a representation (matrix representation) of G with character χ. Suppose that

$$\pi \cong a_1\pi_1 \bigoplus \cdots \bigoplus a_n\pi_n$$

where π_i are pairwise inequivalent irreducible representations (matrix representations) of G. Let χ_i be the character of π_i. Then

- $\chi = a_1\chi_1 + \cdots + a_n\chi_n$
- Suppose ϕ is an irreducible representation (matrix representation) of G with character ψ, then

$$\langle \chi, \psi \rangle_2 = \begin{cases} a_i|G| & \text{if } \phi \cong \pi_i \text{ for some } i \\ 0 & \text{otherwise} \end{cases}$$

  In particular, $\langle \chi, \chi_i \rangle_2 = a_i|G|$

- π is irreducible iff $\langle \chi, \chi \rangle_2 = |G|$

Remark
(3) of this corollary is often the most efficient way to determine whether a representation is irreducible.

Proof

- If $g \in G$, then

$$\chi(g) = \operatorname{tr}(\pi(g)) = \operatorname{tr}(\bigoplus_{i=1}^{n} a_i \pi_i(g))$$
$$= \sum_{i=1}^{n} a_i \operatorname{tr}(\pi_i(g)) = \sum_{i=1}^{n} a_i \chi_i(g)$$

  (trace of big matrix = sum of traces of all diagonal blocks)

- Suppose $\pi : G \to GL(V)$ is a representation. By remark of Def 6.13 and Maschke' Theorem, there exist bases β for V and β_i for subrepresentations s.t.
$$[\pi(g)]_\beta = a_1[\pi_1(g)]_{\beta_1} \bigoplus \cdots \bigoplus a_n[\pi_n(g)]_{\beta_n}$$
  for all $g \in G$. Take the trace for both side
$$\chi(g) = a_1\chi_1(g) + \cdots + a_n\chi_n(g)$$

- Note
$$\langle \chi, \psi \rangle_2 = \langle a_1\chi_1 + \cdots + a_n\chi_n, \psi \rangle_2 = \sum_{i=1}^{n} a_i \langle \chi_i, \psi \rangle_2$$
  The result follows from Theorem 6.25

- From Theorem 6.25
$$\langle \chi, \chi \rangle_2 = \sum_{i=1}^{n} \langle a_i\chi_i, a_i\chi_i \rangle_2 = |G| \sum_{i=1}^{n} a_i^2$$
  Note a_i must be natural numbers, so $\sum_{i=1}^{n} a_i^2 = 1$ iff π is irreducible.

Corollary 6.27. Two representations (matrix representations) of a finite group are equivalent iff they have the same characters.

Remark
Refer to Def. 6.18, where we said a character is irreducible iff the representation producing it is irreducible. This corollary essentially says the decomposition of any representation from Maschke's theorem is unique, up to reording of terms.

Proof
Lemma 6.20 shows equivalent representations (matrix representations) have same characters. Now we show the other direction. We prove for matrix representations, but the proof is the same for representations (since Corollary 6.26 and Maschke's theorem applies to representations as well).

- Let ϕ, π be matrix representations of G with characters χ, ψ. Suppose $\chi(g) = \psi(g)$ for all $g \in G$. By Maschke's theorem,

  $$\phi \cong a_1\phi_1 \bigoplus \cdots \bigoplus a_n\phi_n$$

  for some pairwise inequivalent irreducible matrix representations ϕ_i. Let χ_i be the character of ϕ_i. WLOG, we can assume

  $$\pi \cong b_1\phi_1 \bigoplus \cdots \bigoplus b_n\phi_n$$

  using 0s as coefficients if an irrep occurs in ϕ or π and not in both. By Corollary 6.26,

  $$a_i = \frac{1}{|G|}\langle \chi, \chi_i\rangle_2 = \frac{1}{|G|}\langle \psi, \chi_i\rangle_2 = b_i$$

  for all i. Hence $\phi \cong \pi$

Lemma 6.28. Suppose $\rho : K \to GL(V)$ is a representation of K and χ_ρ is the character. By Lemma 6.17, there exists a basis β of V s.t. $X(k) = [\rho(k)]_\beta$ is a unitary matrix for all k. In particular, $X(k)^{-1} = \overline{X(k)^T}$. Hence

$$\overline{\chi_\rho(k)} = \overline{\mathrm{tr}(X(k))} = \mathrm{tr}(X(k^{-1})^T) = \mathrm{tr}(X(k^{-1})) = \chi_\rho(k^{-1})$$

This gives a simplification on inner products of characters

$$\langle \psi, \chi\rangle_2 = \sum_{k\in K} \psi(k)\chi(k^{-1})$$

6.4 Decomposition of the Right Regular Representation

Main result: decomposition of the (right) regular representation of G as a direct sum of irreps contains **every** irrep of G.

Lemma 6.29. Group G with right regular representation $R : G \to GL(L^2(G))$ by $R(\gamma)f(g) = f(g\gamma)$, character χ_R. Then

$$\chi_R(\gamma) = \begin{cases} |G| & \gamma = e_G \\ 0 & \text{otherwise} \end{cases}$$

Proof
From remark of Def. 1.10, given $G = \{g_1, \ldots, g_n\}$, the standard basis of $L^2(G)$

is $\beta = [\delta_{g_1}, \ldots, \delta_{g_n}]$, where δ_{g_i} is the indicator function of whether $g = g_i$. With respect to this basis, we define χ_R. Then

$$[R(\gamma)]_\beta = \Big([R(\gamma)\delta_{g_1}]_\beta \Big| \cdots \Big| [R(\gamma)\delta_{g_n}]_\beta\Big)$$

Let $x, g \in G$, then

$$(R(\gamma)\delta_g)(x) = \delta_g(x\gamma) = \delta_{g\gamma^{-1}}(x)$$

So $R(\gamma)\delta_g = \delta_{g\gamma^{-1}}$. Then, (g, g) the entry not zero iff $\delta_{g\gamma^{-1}}(g) \neq 0$ iff $g\gamma^{-1} = g$ iff $\gamma = e_G$. Therefore,

$$\chi_R(\gamma) = \mathrm{tr}([R(\gamma)]_\beta) = \sum_{g \in G} \delta_{g\gamma^{-1}}(g) = \begin{cases} |G| & \gamma = e_G \\ 0 & \text{otherwise} \end{cases}$$

Theorem 6.30. Let G be a finite group. Then there are only finitely many irreps of G, up to equivalence. Suppose that $V_1, \ldots, V_n$ form a complete list of inequivalent irreducible representations of G. Let $d_i = dim(V_i)$. Then $L^2(G)$ (right regular representation) is orthogonally equivalent to

$$d_1 V_1 \bigoplus \cdots \bigoplus d_n V_n$$

Moreover, $|G| = d_1^2 + \cdots + d_n^2$

Remark
This result essentially states that right regular representation contains all of the irreducible representations of G. Also, when we talk about right regular representation, we usually consider the standard inner product. Recall right regular representation is unitary with respect to the standard inner product.

Proof
Let χ_R be the character of the right regular representation. Suppose that W is an irreducible representation of G with character χ. Then

$$\langle \chi_R, \chi \rangle_2 = \sum_{g \in G} \chi_R(g)\overline{\chi(g)} = \chi_R(e_G)\overline{\chi(e_G)} = |G| dim(W)$$

Since by Lemma 6.29, $\chi_R(g) = 0$ for $g \neq e_G$. Thus by Corollary 6.26, W occurs in the decomposition of $L^2(G)$ exactly $dim(W)$ times. Then, there must be finitely many irreps of G (worse case: each irreducible representation has dimension 1, then at most $dim(L^2(G))$ such spaces). Hence $L^2(G)$ is orthogonally equivalent to

$$d_1 V_1 \bigoplus \cdots \bigoplus d_n V_n$$

Taking the dimensions of both side, we have $|G| = d_1^2 + \cdots + d_n^2$ (space of dimension d_i appears d_i times).

Remark
A result from group representation theory: the number of irreducible representations of G, up to equivalence, is equal to the number of conjugacy classes of G.

Corollary 6.31. Finite group G, $V_1, \ldots, V_n$ complete set of inequivalent irreps of G. Suppose V_1 is the trivial representation of G (that is, $\phi : G \to GL(\mathbb{C})$). Let $d_i = dim(V_i)$. Then $L_0^2(G)$ is orthogonally equivalent to $d_2V_2 \bigoplus \cdots \bigoplus d_nV_n$.

Remark
Recall that $L_0^2(G)$ is used to calculate the second-largest eigenvalue of a Cayley graph. Here we show it corresponds to non-trivial irreducible representations of the group.

Proof
Suffice to show that $L_0^2(G)^\perp = V_1$. Let f_0 be the function that is constant to 1 on G. Note

$$\mathbb{C}f_0 = \{f \in L^2(G) \,|\, f \text{is a constant function}\}$$

and

$$L_0^2(G) = \{f \in L^2(G) \,|\, \langle f, f_0 \rangle_2 = 0\}$$

So $L_0^2(V)^\perp = \mathbb{C}f_0$, and $L^2(G) = \mathbb{C}f_0 \bigoplus L_0^2(V)$. Also, for $f \in \mathbb{C}f_0$ and $\gamma, g \in G$, $(R(\gamma)f)(g) = f(g\gamma) = constant = f(g)$, thus $\mathbb{C}f_0$ corresponds to the trivial representation. Done.

6.5 Uniqueness of Invariant Inner Product

This section shows that there is essentially only one invariant inner product (up to a scalar constant) on an irrep. Recall in Lemma 6.14 we showed that every representation has at least one invariant inner product.

Definition 6.32. Let V be a vector space. Let $V^* = \{\phi : V \to \mathbb{C} \,|\, \phi \text{ is linear}\}$, the **dual space** of V. Define operations on V^* using all vectors in V:

- Addition: $(\phi_1 + \phi_2)(v) = \phi_1(v) + \phi_2(v)$
- Scalar multiplication: $(c\phi)(v) = c\phi(v)$

Dual space V^* is a vector space over $\mathbb{C}$.

Definition 6.33. Let V be a representation for G, V^* be the dual space of V. Define an action of G on V^* as follows. For $\phi \in V^*$, define $g \cdot \phi \in V^*$ by $(g \cdot \phi)(v) = \phi(g^{-1} \cdot v)$. With this action, call V^* the dual representation of V.

Proof
To verify that V^* is a representation, suffice to show the action has $g \cdot h \cdot \phi = gh \cdot \phi$ for any $g, h \in G, \phi \in V^*$ (thus the corresponding map is homomorphism). For any $v \in G$

$$g \cdot (h \cdot \phi)(v) = h \cdot \phi(g^{-1} \cdot v) = \phi(h^{-1} \cdot g^{-1} \cdot v) = \phi((gh)^{-1} \cdot v) = gh \cdot \phi(v)$$

Definition 6.34.

- Given a G-invariant inner product $\langle \cdot, \cdot \rangle$ on a representation V and $v \in V$, define $H_v : V \to \mathbb{C}$ by $H_v(\cdot) = \langle \cdot, v \rangle$
- Note H_v is linear, so $H_v \in V^*$. Define $H : V \to V^*$ by $H(v) = H_v$.
- A, B vector spaces over $\mathbb{C}$, $f : A \to B$. f is conjugate-linear (semilinear) if $f(c_1 v_1 + c_2 v_2) = \overline{c_1} f(v_1) + \overline{c_2} f(v_2)$ for all $c_1, c_2 \in \mathbb{C}, v_1, v_2 \in A$

Lemma 6.35. The map H is

- bijective
- conjugate-linear
- G-invariant

Proof

- Let $[v_1, \ldots, v_n]$ be an orthonormal basis for V. Assume $H(v) = H(u)$, then $H_v(w) = H_u(w)$ for any $w \in V$. We have $v = a_1 v_1 + \ldots a_n v_n, \quad u = b_1 v_1 + \cdots + b_n v_n$, so $\overline{a_j} = \langle v_j, v \rangle = H_v(v_j) = H_u(v_j) = \langle v_j, u \rangle = \overline{b_j}$ for all j, thus $v = u$ injective done.
- For arbitrary $\phi \in V^*$, let $a_j = \phi(v_j)$, let $v = \overline{a_1} v_1 + \cdots + \overline{a_n} v_n$. Then $H_v(v_j) = \phi(v_j)$ for all j. Since two linear maps agree on basis, they agree everywhere, thus $H(v) = H_v = \phi$
- Definition of inner product.
- Note G-invariant inner product $\langle \cdot, \cdot \rangle$. For all $g \in G, v, w \in V$, we have

$$\langle g^{-1} \cdot w, v \rangle = \langle w, g \cdot v \rangle$$

definition of invariant

$$H_v(g^{-1} \cdot w) = H_{g \cdot v}(w)$$

definition of H_v

$$(g \cdot H_v)(w) = H_{g \cdot v}(w)$$

since $H_v \in V^*$ and the action defines a representation

$$g \cdot H_v = H_{g \cdot v}$$

$$g \cdot (H(v)) = H(g \cdot v)$$

Definition 6.36. Suppose G-invariant inner product $\langle\cdot,\cdot\rangle$, $\langle\cdot,\cdot\rangle'$ on V, define H, H' following Def. 6.34, let $\psi = H^{-1} \circ H'$. Then ψ is a G-isomorphism.

Proof
Bijectivity is trivial. Suffice to show H^{-1} is conjugate-linear and G-invariant. Let $f_1, f_2 \in V^*$, let $v_1 = H^{-1}(f_1), v_2 = H^{-1}(f_2)$, then

$$\begin{aligned}
H^{-1}(c_1 f_1 + c_2 f_2) &= H^{-1}(c_1 H(v_1) + c_2 H(v_2)) \\
&= H^{-1}(H(\overline{c_1} v_1 + \overline{c_2} v_2)) \\
&= \overline{c_1} v_1 + \overline{c_2} v_2 \\
&= \overline{c_1} H^{-1}(f_1) + \overline{c_2} H^{-1}(f_2)
\end{aligned}$$

Conjugate-linear done. Let $f \in V^*, g \in G$, let $t = H^{-1}(f)$ then

$$H^{-1}(g \cdot f) = H^{-1}(g \cdot H(t)) = H^{-1} H(g \cdot t) = g \cdot t = g \cdot H^{-1}(f)$$

G-invariant done.

Proposition 6.37. Suppose V is an irrep for a finite group G, G-invariant inner product $\langle\cdot,\cdot\rangle$, $\langle\cdot,\cdot\rangle'$ on V. Then $\langle\cdot,\cdot\rangle = C\langle\cdot,\cdot\rangle'$ for some $C \in \mathbb{R}^+$. That is, $\langle v, w\rangle = C\langle v, w\rangle'$ for all $v, w \in V$.

Proof

6.6 Induced Representations

Definition 6.38. Let G be a finite group and $H \subset G$, $\sigma : H \to GL(W)$ be a representation of H. Define the vector space

$$V = \{f : G \to W \mid f(hg) = \sigma(h)f(g) \text{ for all } g \in G, h \in H\}$$

Let $\phi : G \to GL(V)$ be defined as $(\phi(x)f)(g) = f(gx)$. The map is the **induced representation** from H up to G, denoted by $Ind_H^G(\sigma)$.

Proof

$$(\rho(x)\rho(y)f)(g) = \rho(y)f(gx) = f(gxy) = (\rho(xy)f)(g)$$

Homomorphism done, indeed representation.

Remark
Since $e \in H \subset G$, $f(h) = \sigma(h)f(e) = \sigma(h)$. That is, the induced representation is the same with σ on H. We are extending the representation of H to get a representation of G.

Definition 6.39. Same definitions as Def. 6.38,

- By Lemma 6.14, exists inner product $\langle\cdot,\cdot\rangle_W$ with respect to which σ is unitary.
- Let $g_1,\dots,g_m$ form a set of representatives for $H\backslash G$ (set of all right coset Hg), exists an orthonormal basis $e_1,\dots,e_s$ of W with respect to $\langle\cdot,\cdot\rangle_W$.
- Let
$$\tilde{\sigma}(y)=\begin{cases}\sigma(y) & y\in H\\ 0 & y\notin H\end{cases}$$
For each $i=1,\dots,m, j=1,\dots,s$ define
$$f_{ij}(x)=\tilde{\sigma}(xg_i^{-1})e_j$$

Remark
Somehow this is extending an orthonormal basis of W corresponding to H to an orthonormal basis of V corresponding to G.

Proof
Discuss $gg_i^{-1}\in H$ and $gg_i^{-1}\notin H$, easy to show $f_{ij}\in V$.

Remark
Give $x=hg$, we have $f_{ij}(x)=\sigma(h)\hat{\sigma}(gg_i^{-1})e_j=\hat{\sigma}(h)\hat{\sigma}(gg_i^{-1})e_j$

Definition 6.40. Following previous definitions,

$$\langle y_1,y_2\rangle_V=\sum_{t=1}^{m}\langle y_1(g_t),y_2(g_t)\rangle_W$$

is a well-defined $G-$invariant inner product on V.

Remark
We extend the H-invariant inner product on W to an inner product on V, and the extended version is G-invariant on V as well.

Proof
We first show the inner product does not depend on the choice of representatives (hence well-defined). Suppose the other representatives of $H\backslash G$ is $g_1',\dots,g_m'$, then $h_1g_1'=g_1,\dots,h_mg_m'=g_m$ for some $h_i\in H$. Let $y_1,y_2\in V$, then

$$\begin{aligned}\sum_{t=1}^{m}\langle y_1(g_t),y_2(g_t)\rangle_W&=\sum_{t=1}^{m}\langle y_1(h_tg_t'),y_2(h_tg_t')\rangle_W\\&=\sum_{t=1}^{m}\langle\sigma(h_t)y_1(g_t'),\sigma(h_t)y_2(g_t')\rangle_W\\&=\sum_{t=1}^{m}\langle y_1(g_t'),y_2(g_t')\rangle_W\end{aligned}$$

Since σ is unitary with respect to $\langle\cdot,\cdot\rangle_W$. $\langle\cdot,\cdot\rangle_V$ is obviously an inner product.

We now show $\langle\cdot,\cdot\rangle_V$ is G-invariant. Let $f_1, f_2 \in V$. Let $x = hg \in G$. For each t, $g_t x = g_t hg = h_t \hat{g}_t$ for some $h_t \in H, \hat{g}_t \in \{g_1, \ldots, g_m\}$, since $g_t hg$ is also an element of G. Then

$$\begin{aligned}
\hat{g}_{t_1} &= \hat{g}_{t_2} \\
h_{t_1}^{-1}{}_t g_{t_1} hg &= h_{t_2}^{-1}{}_t g_{t_2} hg \\
h_{t_2} h_{t_1}^{-1}{}_t g_{t_1} &= g_{t_2} \\
g_{t_1} &= g_{t_2}
\end{aligned}$$

since g_t are representatives of cosets. That is, $\hat{g}_{t_1} = \hat{g}_{t_1}$ iff $t_1 = t_2$, there is a bijection between g_t and $\hat{g}_t$, summing over all g_t is equivalent to summing over all $\hat{g}_t$. So

$$\begin{aligned}
\langle\rho(x)f_1, \rho(x)f_2\rangle_V &= \sum_{t=1}^{m} \langle\rho(x)f_1(g_t), \rho(x)f_2(g_t)\rangle_W \\
&= \sum_{t=1}^{m} \langle f_1(g_t x), f_2(g_t x)\rangle_W = \sum_{t=1}^{m} \langle\sigma(h_t)f_1(\hat{g}_t), \sigma(h_t)f_2(\hat{g}_t)\rangle_W \\
&= \sum_{t=1} \langle f_1(g_t), f_2(g_t)\rangle_W = \langle f_1, f_2\rangle_V
\end{aligned}$$

Lemma 6.41. The functions f_{ij} form an orthonormal basis for V.

Remark
Since $i = 1, \ldots, m$ is the number of right coset representatives of $H\backslash G$, $j = 1, \ldots, s$ corresponds to an orthonormal basis of W, we have $dim(V) = dim(W)[G : H]$.

Proof
Note $\tilde{\sigma}(g_a g_b^{-1}) \neq 0$ iff $g_a g_b^{-1} \in H$ iff $a = b$, since g_t are representatives of cosets. Suppose $1 \leq i, k \leq m, 1 \leq j, r \leq s$, then

$$\begin{aligned}
\langle f_{ij}, f_{kr}\rangle_V &= \sum_{t=1}^{m} \langle f_{ij}(g_t), f_{kr}(g_t)\rangle_W \\
&= \sum_{t=1}^{m} \langle\tilde{\sigma}(g_t g_i^{-1})e_j, \tilde{\sigma}(g_t g_k^{-1})e_r\rangle_W \\
&= \begin{cases} 1 & i = k,\ j = r \\ 0 & \text{otherwise} \end{cases}
\end{aligned}$$

Hence, f_{ij} are mutually orthogonal and are each of norm 1, thus they form a linearly independent set in V. Suffice to show they also span V.

Let $f \in V, x = hg_r \in G$. Recall in Def. 6.39, $e_1, \ldots, e_s$ is an orthonormal basis of W, V is the vector space of function from G to W. Suppose $f(g_r) = c_1e_1 + \cdots + c_se_s$, then $\langle f(g_r), e_1\rangle_W = c_1$ since this is a set of orthonormal basis. Note $f_{ij}(x) = \tilde{\sigma}(hg_rg_i^{-1})e_j$, which simplifies to $\sigma(h)e_j$ if $i = r$, or to 0 if $i \neq r$. Thus,

$$\begin{aligned}
f(x) &= \sigma(h)f(g_r) = c_1\sigma(h)e_1 + \cdots + c_s\sigma(h)e_s \\
&= \langle f(g_r), e_1\rangle_W \sigma(h)e_1 + \cdots + \langle f(g_r), e_s\rangle_W \sigma(h)e_s \\
&= \langle f(g_r), e_1\rangle_W f_{r1}(x) + \cdots + \langle f(g_r), e_s\rangle_W f_{rs}(x) \\
&= \sum_{b=1}^{s} \langle f(g_r), e_b\rangle_W f_{rb}(x)
\end{aligned}$$

Hence

$$f = \sum_{a=1}^{m}\sum_{b=1}^{s} \langle f(g_a), e_b\rangle_W f_{ab}$$

since with $x = hg_t$, for $r \neq t$, $\sum_{b=1}^{s}\langle f(g_r), e_b\rangle_W f_{rb} = 0$. That is, f_{ij} span V.

Proposition 6.42. Let $G, H, \sigma, V, \rho = Ind_H^G(\sigma)$ be defined as before, let

$$\tilde{\chi}_\sigma(x) = \begin{cases} \chi_\sigma(x) & x \in H \\ 0 & x \notin H \end{cases}$$

where χ_σ is the character of σ. Let χ_ρ be the character of ρ. Then

$$\chi_\rho(g) = \frac{1}{|H|}\sum_{x \in G} \tilde{\chi}_\sigma(xgx^{-1})$$

Remark
Finally, we extend the character for the representation of H to the character for the representation of G, and we are ready to build connection between these two representations using the close relation between character and representation.

Proof
Let $x \in G$, then $x = hg_k$ for some $h \in H, 1 \leq k \leq m$. Then

$$(\rho(g)f_{ij})(x) = f_{ij}(hg_kg) = \tilde{\sigma}(h)\tilde{\sigma}(g_kgg_i^{-1})e_j$$

Denote the orthonormal basis $[e_1, \ldots, e_s]$ as β_W, let

$$[\tilde{\sigma}(x)]_{\beta_W} = \begin{pmatrix} \tilde{\sigma}_{11}(x) & \cdots & \tilde{\sigma}_{1s}(x) \\ \vdots & \ddots & \vdots \\ \tilde{\sigma}_{s1}(x) & \cdots & \tilde{\sigma}_{ss}(x) \end{pmatrix}$$

That is, we defined coordinate functions for the matrix form of $\tilde{\sigma}$, and we have $\tilde{\sigma}(x)e_j = \sum_{r=1}^{s} \tilde{\sigma}_{rj}(x)e_r$. That is, $\tilde{\sigma}(x)$ acts on e_j is the jth column of the above matrix, exactly how we get the matrix form with respect to this basis.

Since $h = xg_k^{-1}$, we have

$$(\rho(g)f_{ij})(x) = \tilde{\sigma}(h)\sum_{r=1}^{s} \tilde{\sigma}_{rj}(g_k g g_i^{-1})e_r = \sum_{r=1}^{s} \tilde{\sigma}_{rj}(g_k g g_i^{-1})\tilde{\sigma}(xg_k^{-1})e_r = \sum_{r=1}^{s} \tilde{\sigma}_{rj}(g_k g g_i^{-1})f_{kr}(x)$$

Note $f_{ar}(x) = \tilde{\sigma}(hg_k g_a^{-1})e_r \neq 0$ iff $a = k$, hence

$$\rho(g)f_{ij} = \sum_{a=1}^{m}\sum_{r=1}^{s} \tilde{\sigma}_{rj}(g_a g g_i^{-1})f_{ar}$$

Let β_V be the ordered basis $[f_{11}, \ldots, f_{1s}, f_{21}, \ldots, f_{2s}, \ldots, f_{m1}, \ldots, f_{ms}]$, then

$$[\rho(g)]_{\beta_V} = \begin{pmatrix} [\tilde{\sigma}(g_1 g g_1^{-1})]_{\beta_W} & \cdots & [\tilde{\sigma}(g_1 g g_m^{-1})]_{\beta_W} \\ \vdots & \ddots & \vdots \\ [\tilde{\sigma}(g_m g g_1^{-1})]_{\beta_W} & \cdots & [\tilde{\sigma}(g_m g g_m^{-1})]_{\beta_W} \end{pmatrix}$$

The matrix is written in block form, with each block as what we defined above for W. For example, $\rho(g)f_{11}$ corresponds to the first column of this matrix. With this matrix, we have

$$\operatorname{tr}([\rho(g)]_{\beta_V}) = \sum_{r=1}^{m} \operatorname{tr}(\tilde{\sigma}(g_r g g_r^{-1})) = \sum_{r=1}^{m} \tilde{\chi}_\sigma(g_r g g_r^{-1})$$

Since $\tilde{\chi}_\sigma(t) = 0$ iff $t \notin H$ iff $\tilde{\sigma}(t) = 0$. With $x = hg_r$, $\tilde{\chi}_\sigma(g_r g g_r^{-1}) = \tilde{\chi}_\sigma(hg_r g g_r^{-1}h^{-1}) = \tilde{\chi}_\sigma(xgx^{-1})$ (same character under conjugation), thus

$$|H|\sum_{r=1}^{m} \tilde{\chi}_\sigma(g_r g g_r^{-1}) = \sum_{x\in G} \tilde{\chi}_\sigma(xgx^{-1})$$

On the RHS, we sum over G, correspondingly we sum each representative g_r $|H|$ times. Hence,

$$\chi_\rho(g) = \operatorname{tr}([\rho(g)]_{\beta_V}) = \frac{1}{|H|}\sum_{x\in G} \tilde{\chi}_\sigma(xgx^{-1})$$

Definition 6.43. Finite group G, $H \subset G$. Let $\phi : G \to GL(V)$ be a representation of G. Define the **restriction** of ϕ to H as the representation $Res_H^G(\phi) : H \to GL(V)$, where $Res_H^G(\phi)(h) = \phi(h)$ for all $h \in H$.

Theorem 6.44. (Frobenius Reciprocity)
Finite group G, $H \subset G$, $\sigma : H \to GL(V_1)$, $\phi : G \to GL(V_2)$ two representations. Let $\phi = Ind_H^G(\sigma), \tau = Res_H^G(\phi)$, then

$$\frac{1}{|G|}\langle \chi_\rho, \chi_\phi \rangle_2 = \frac{1}{|H|}\langle \chi_\sigma, \chi_\tau \rangle_2$$

where both inner products are standard.

Proof
By Prop. 6.42, and the fact that character is constant on conjugacy class (Lemma 6.20)

$$\begin{aligned}
\frac{1}{|G|}\langle \chi_\rho, \chi_\phi \rangle_2 &= \frac{1}{|G||H|}\sum_{g\in G}\sum_{x\in G}\tilde{\chi}_\sigma(xgx^{-1})\overline{\chi_\phi(g)} \\
&= \frac{1}{|G||H|}\sum_{g\in G}\sum_{x\in G}\tilde{\chi}_\sigma(xgx^{-1})\overline{\chi_\phi(xgx^{-1})} \\
&= \frac{1}{|G||H|}\sum_{x\in G}\sum_{y\in G}\tilde{\chi}_\sigma(y)\overline{\chi_\phi(y)} \\
&= \frac{1}{|H|}\sum_{y\in G}\tilde{\chi}_\sigma(y)\overline{\chi_\phi(y)} \\
&= \frac{1}{|H|}\sum_{h\in H}\chi_\sigma(h)\overline{\chi_\tau(h)} \\
&\text{since } \tilde{\chi}_\sigma \text{ is 0 if not in } H \\
&= \frac{1}{|H|}\langle \chi_\sigma, \chi_\tau \rangle_2
\end{aligned}$$

Remark
Main usage of this result is to check which irreps appear in the direct sum decomposition of the induced representation.

Example
$H \subset G$, suppose exists σ as a nontrivial irrep of H. Let $\rho = Ind_H^G(\sigma)$, 1 be "the" trivial representation of G (hence "the" trivial representation of H). Since 1 is always an irrep and not equivalent to σ, we have $\langle 1, \chi_\sigma \rangle = 0$ (by Theorem 6.25). Then, by Frobenius Reciprocity,

$$\frac{1}{|G|}\langle 1, \chi_\rho \rangle = \frac{1}{|H|}\langle 1, \chi_\sigma \rangle = 0$$

Hence, by Corollary 6.26, the trivial representation cannot appear in the direct sum decomposition of ρ.